

ELEC 442

Lab 0

GOALS

- ☐ *Install Core Software*
- ☐ *Install Quanser Software*
- ☐ *Test Software*

1 Environment Setup

The following guides in the Lab 0 repo give detailed instructions. **If you need help please ask an instructor/TA. For Lab 0, no submission is required.**

1. [Software Installation Guide \(personal_software_installation.md\)](#)
2. [Computer Setup Guide \(personal_computer_setup.md\)](#)

1.1 Software Setup (All Students)

The following software is required for the labs: Git, Python, and VSCode need to be setup.

1.2 Virtual Arm Setup (Windows only)

In this course, you may also want to work with a Virtual QArm (a digital twin running in Quanser Interactive Labs) alongside the physical hardware. Having this set up on your personal computer enables you to test code at home, experiment safely, and collaborate with your lab partners in parallel.

Unfortunately, QLABS is windows only for the QArm. If you have access to a windows machine, it is strongly recommended, however due to possible incompatibilities, it will not be required to complete the lab.

1.3 Setup Checklist

Before walking into the lab, ensure you have:

- [] Installed core software
 - [] Git
 - [] Python
 - [] VS Code

Optional Extras:

- [] Installed QLABS and created your Quanser account.

2 Environment Test

To verify the setup, a few test scripts have been provided. Clone the Lab-0 Repo to your computer and proceed.

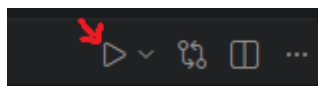
2.1 Core Software Test

2.1.1 Git and VS Code

Try cloning the Lab 0 GitHub Repo to your device. Then try opening it in VS Code. Ask instructors for help if issues arise.

2.1.2 Python

With Lab 0 open in VS Code, open `Lab_0-python_test.py` and try to run it. This can be done by pressing the “play” button in the top right of VS Code (see below)



A terminal pane should pop out and display the following output (numpy and python versions could cause the number to differ slightly, but it should be ≈ 0):

```
hello world
1.2246467991473532e-16
```

Alternatively, you can open a terminal and enter:

```
python replace_this_with_path_to_Lab_0-python_test.py
```

2.2 Q Labs/Virtual Arm Test

1. Open “Quanser Interactive Labs”
2. Sign in
3. Select “QArm” > “QArm Workspace”
4. Run `Lab_0-QLabs_test.py` from the Lab 0 Repo

If successful, you should see the virtual QArm move between an outstretched position and a neutral position every three seconds for 10 cycles.

Congrats! You now have all required software setup, and you’re ready for the rest of the labs!

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